Requirements in Space Missions

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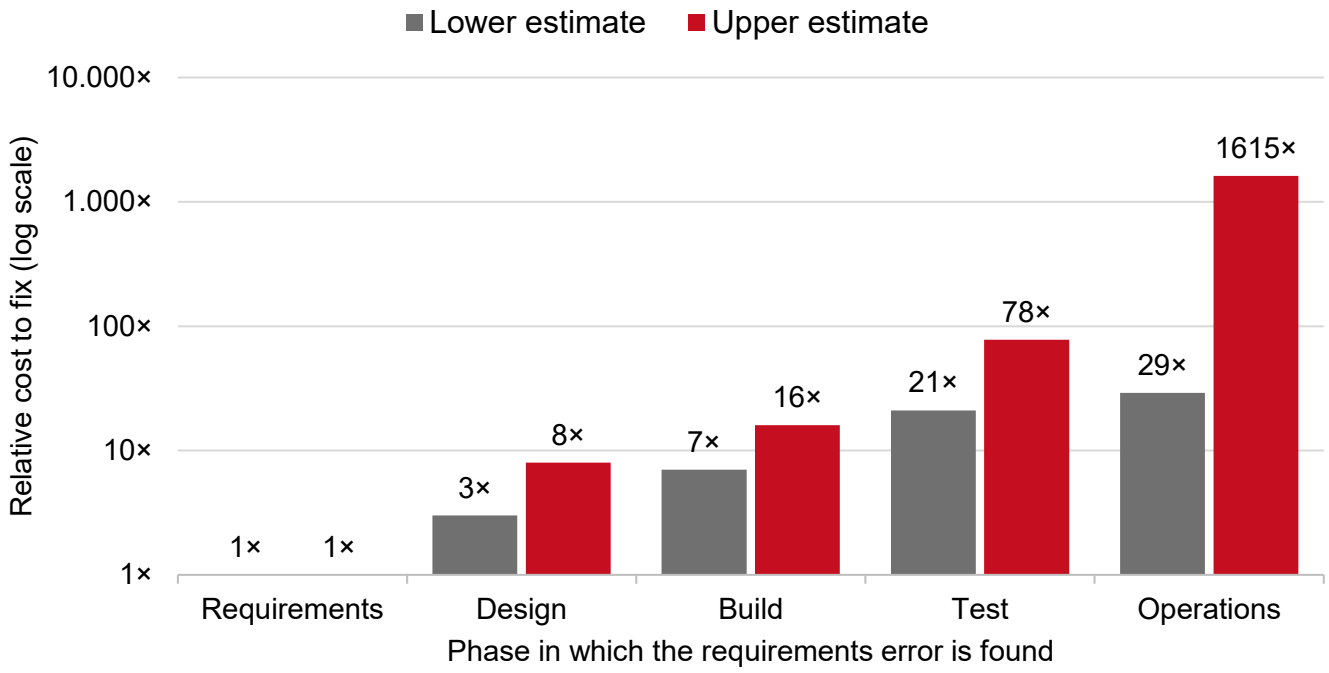
Introduction

Why requirements?

- Means of common ground between user/stakeholder and developers
- Baseline for system verification and testing
- Aims to reduce later need for re-design
- Helps to estimate development effort, time and cost

Introduction

Why requirements? – Errors are cheapest to fix where they are made [3]



System-level cost factors from three estimation methods, Stecklein et al. (2004), Table 13 [3]

- Error introduced in the requirements phase,
  cost to fix = 1× if found there
- Found in test: **21–78×**
- Found in operations: **29× to >1500×**

➔ invest early in requirements and reviews

Introduction

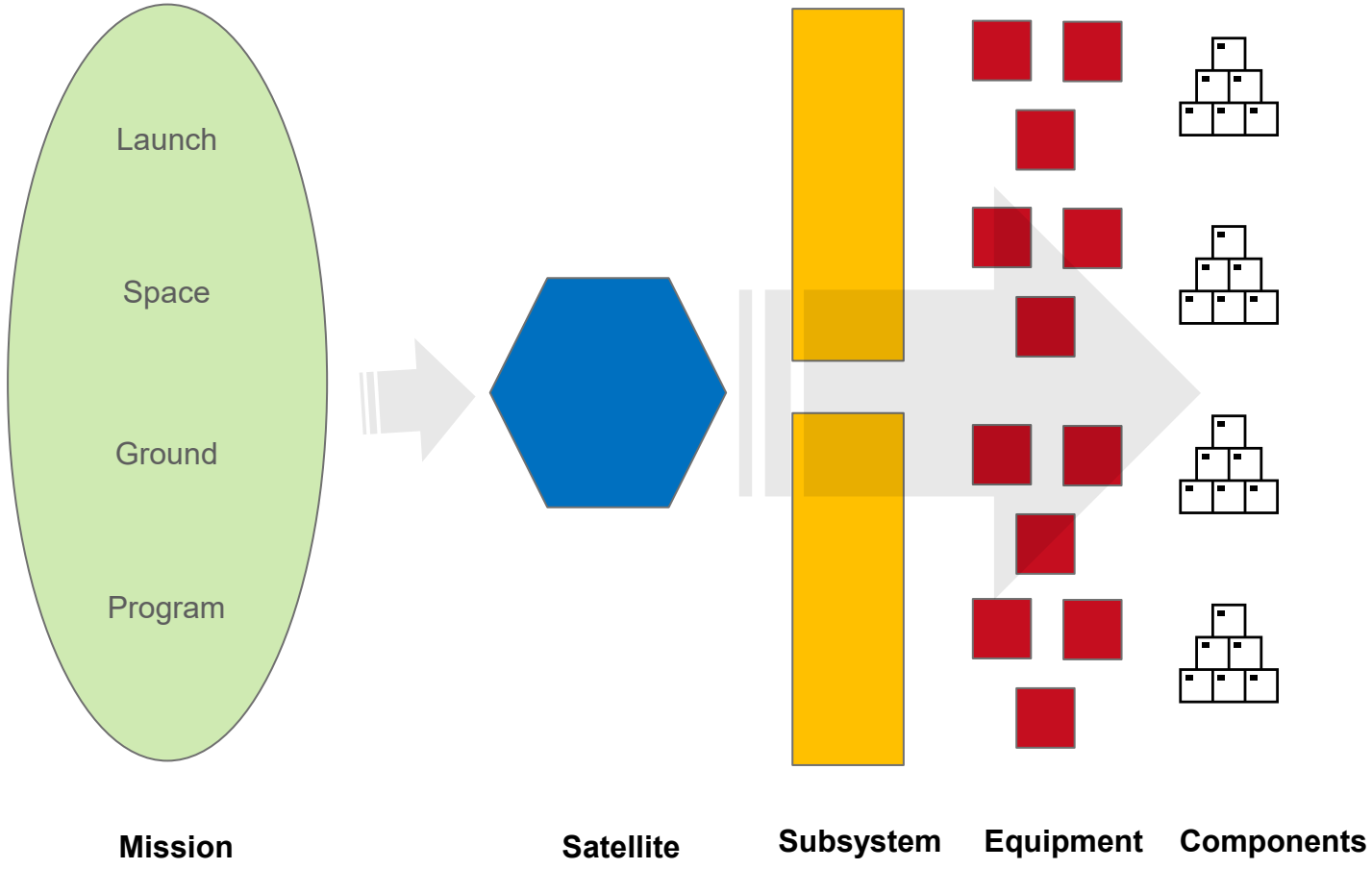
- Requirements are defined at the beginning of each mission

  ➔ iterative top-down process
- Key is to find the balance between too many and too few requirements
- Most requirements are interconnected

  ➔ defining correct relations between requirements important
- There are various approaches on how to derive requirements

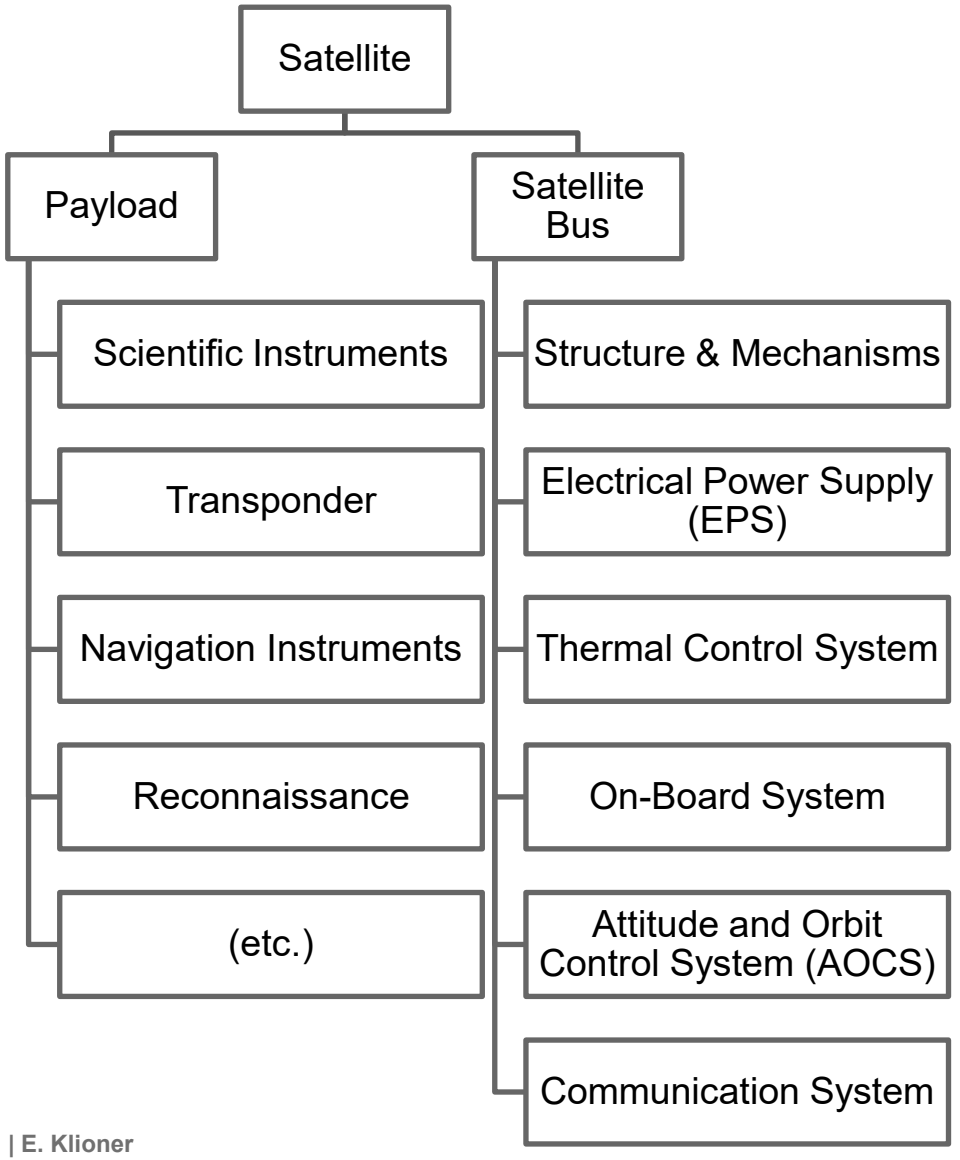
Introduction

- Requirements defined on different levels:
  - Mission level
  - System level
  - Sub-system level
- Requirements touch all satellite subsystems and components



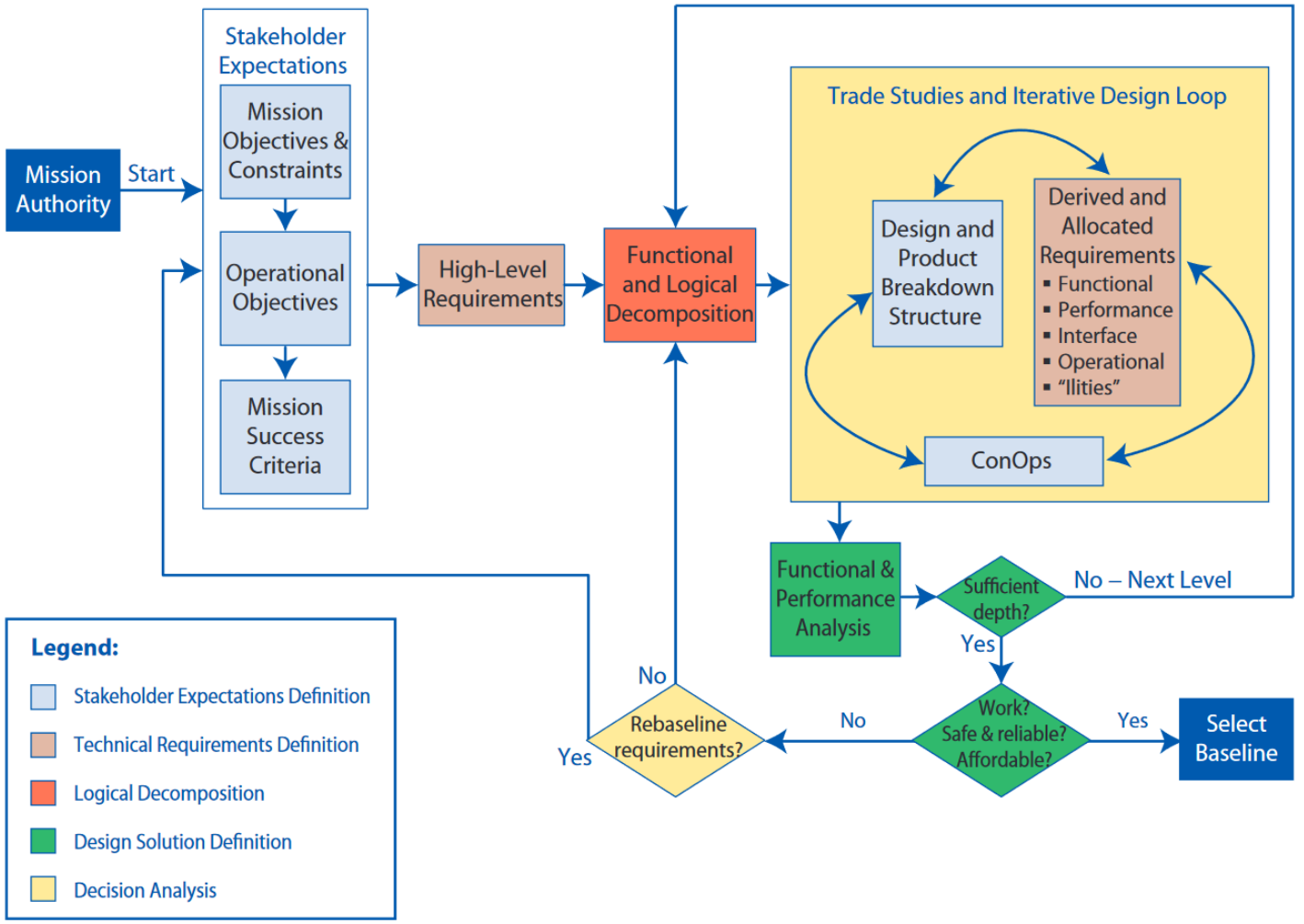
Introduction

Satellite: Bus + Payload



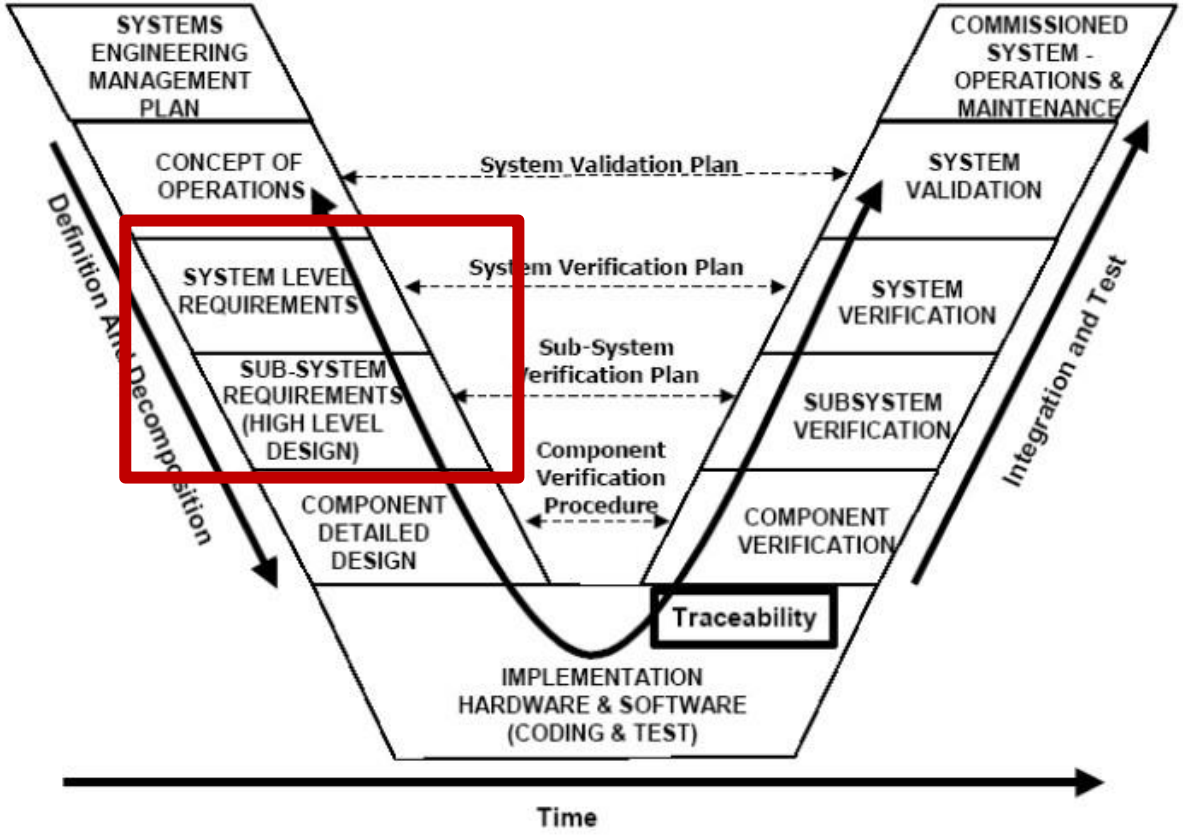
Requirements Definition Process

System Design Process



[1]

V-Model of Systems Engineering



[2]

Top-Down Approach

- Start with top of hierarchy: top-level functions
- Dependencies of each top-level requirement is identified

  ➔ what is affected by each requirement
- Once “bottom” is reached: iteration loop back to the top

  ➔ component-requirement cannot interfere with top-level!
- Trades can be made on the way, but only on the same level
- Interaction/check-ups with stakeholders

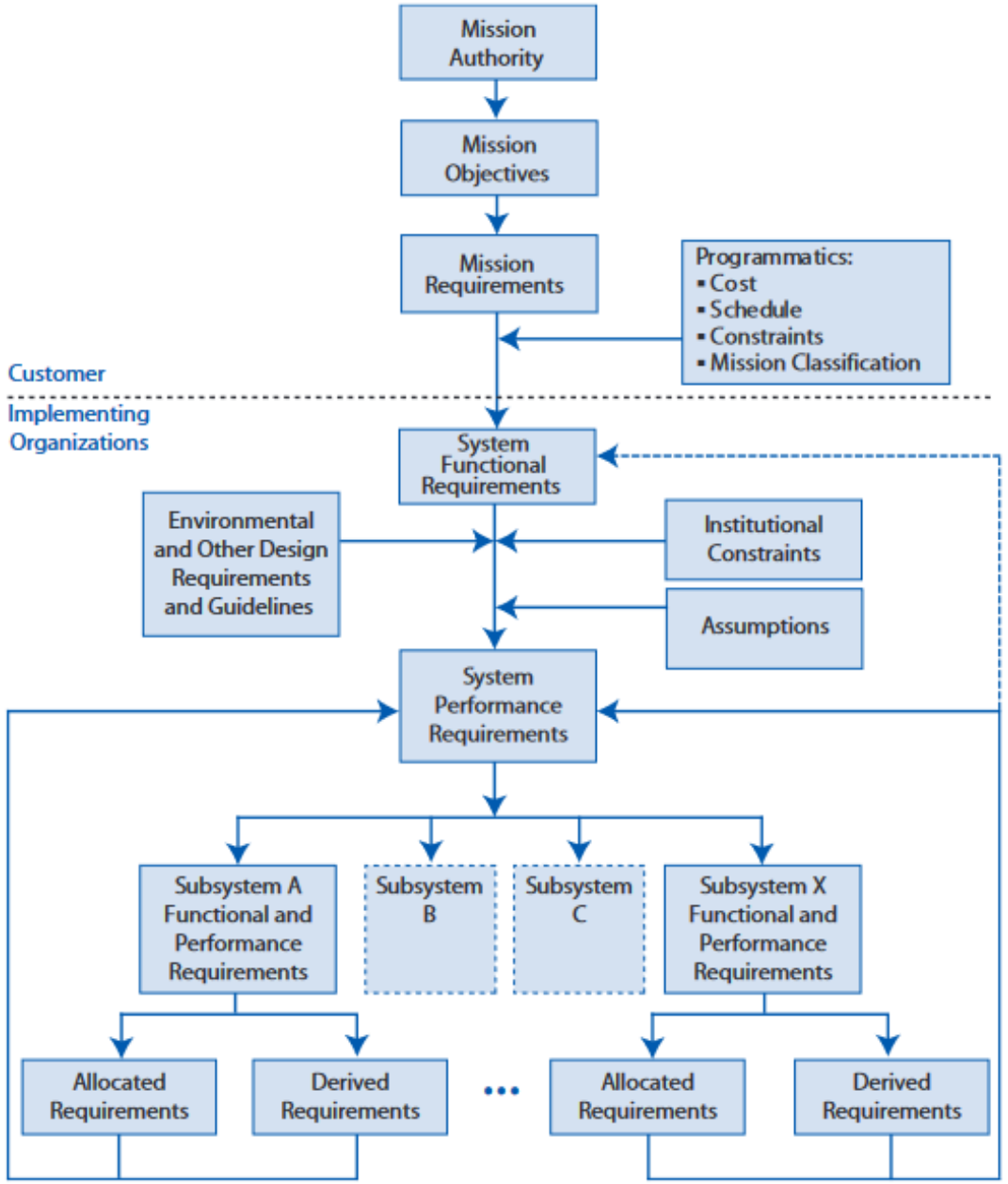
Requirements Analysis

- Decomposing each requirement into lower-level fragments

  ➔ definition of lower-level functions and setting performance criteria for them

- Allocating each new function to corresponding lower-level design elements

Requirements flowdown [1]



Important to distinguish between allocated and derived requirements!

Requirements Documentation and Traceability (ECSS)

Mission Statement (MS)
Document expressing the set of collected needs – customer, Phase 0 [4]

→

Mission Description Document (MDD)
Describes a concept meeting the preliminary TS – one per candidate concept, Phase 0/A [5]

→

Technical Requirements Specification (TS)
Requirements baseline of the product – released at PRR, updated at SRR [6], [10]

→

Specification tree
Hierarchical relationship of all TS for the elements of the system (subsystem, equipment) – released at PDR [10]

Managed in parallel along the whole flow-down:

Interfaces
Interface Requirements Document (IRD) → Interface Control Document (ICD) [7]

Traceability
Requirements Traceability Matrix (RTM): links each requirement to its higher-level requirement(s) [5]

Verification
Verification Control Document (VCD): method, level, stage and status of every requirement [8]

Requirements Budgeting

- Necessary to define not only absolute values but also margins of error and performance budgets
- Based on feasibility and mission objectives
- Used for trade-offs and later verification
- Can be done for all requirement levels allowing for a quantitative budget

Requirements Budgeting – Example: ESA Margin Philosophy [9]

Margins required in ESA science assessment studies (excerpt)

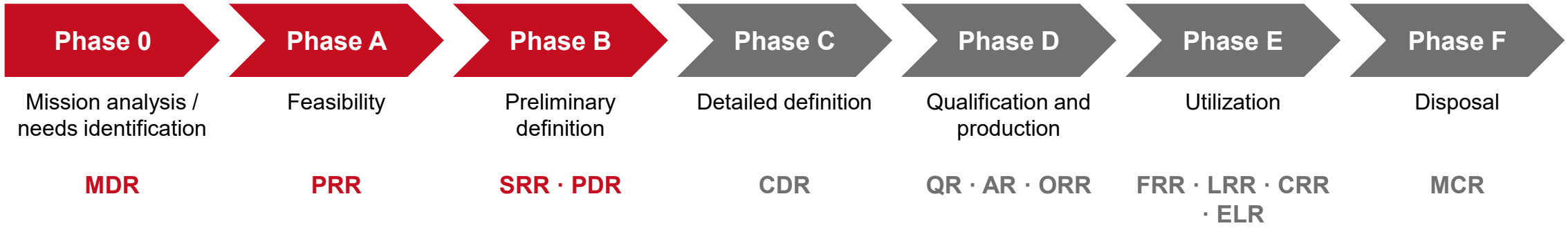
Budget	Level	Margin
Mass	Equipment	≥ 5 % off-the-shelf · ≥ 10 % off-the-shelf with minor modifications · ≥ 20 % new design or major modification / re-design
Mass	System	20 % of the nominal dry mass at launch (nominal dry mass already includes the equipment margins)
Δv	Manoeuvres	5 % for accurately calculated manoeuvres · 100 % for general orbit maintenance and for attitude control / momentum management
Power	Equipment / System	5 / 10 / 20 % by design maturity (conventional electronic units) · ≥ 20 % system margin on nominal power
Data handling	On-board	≥ 50 % memory margin (RAM) · processor peak usage ≤ 50 % of maximum capability

➔ Margin policy is agreed between customer and supplier, as a function of technology readiness [5]

System Requirements Review (SRR)

- Internal review where all requirements are being analyzed in terms of completeness, correctness, logic, etc.
- ECSS: During phase B; NASA: Typically end of phase A, before phase B
- Goal: min. top 3 requirement levels must be defined, validated and documented
- Use of configuration management

Requirements in the Project Life Cycle (ECSS) [10]



- **MDR** – release of the mission statement; assessment of the preliminary technical requirements specification (TS)
- **PRR** – release of the TS; confirmation of technical and programmatic feasibility of the system concept(s)
- **SRR** – release of the updated TS; assessment of the preliminary design and verification programme
- **PDR** – verification of the preliminary design; release of product tree, WBS and specification tree

Types of Requirements

Types of requirements

Top-Level Requirements

- Functional
  - What function must the system have to accomplish mission objectives?
  - E.g. performance, coverage, responsiveness
- Operational
  - How does the system operate, how do users interact with it?
  - E.g. duration, availability, life span, data handling/distribution

Types of requirements

Subsystem or Component Requirements

- Also structured hierarchically
- Functional and operational requirements
- Performance requirements

  ➔ quantitative requirements of a performance characteristic, e.g. Pointing of AOCS
- Verification requirements
- Documentation requirements

Types of Requirements

Further Technical Requirements Types (ECSS-E-ST-10-06C) [6]

- Interface req.
- Environmental req.
- Human factor req.
- Logistics support req.
- Physical req.
- Product assurance induced req.
- Configuration req.
- Design req.

Constraints

➔ No “real” requirements

➔ Not derived from mission objectives

- But: Some sources name them however as requirements that cannot be traded
- E.g. cost, schedule, laws, politics, interfaces

Requirements: Dos and Don’ts

Dos (1)

- Clear and consistent
- Correct
- Feasible
- Flexibility
- Without ambiguity
- Singular
- Verify

Dos (2)

For sets of requirements:

- Absence of redundancy
- Consistency
- Completeness
- Absence of conflicts

Don’ts

- Use verifiable requirements

  ➔ no “maximize”, “minimize”, “fast/rapid”, “easy”, “sufficient” etc.
- Write “how” instead of “what”
- Don’t use “support”, “but not limited to”, “etc”, “and/or”
- Don’t overspecify
- Check for grammar/spelling, general logic

Anatomy of a Requirement

Structure of a well-formed requirement (illustrative example)

SYS-EPS-030	The EPS	**shall**	provide	≥ 60 W orbit-average power to the payload	in Science Mode at end of life (EOL).
Unique ID	Subject	Verbal form	Action	Performance (value, unit)	Condition

Attributes managed with each requirement

- Unique identifier
- Rationale
- Parent (trace)
- Verification method
- Responsible entity
- Status (open values, TBD)

ECSS-E-ST-10-06C, cl. 8.2: each technical requirement is quantified, justified, traceable, unambiguous, unique, identifiable, singular, self-contained, verifiable and has a specified tolerance [6]

Wording of Requirements

Verbal forms [6], [11]

- “shall” = requirement · “should” = recommendation (NASA: goal)
- “may” = permission · “can” = possibility or capability
- NASA: “will” = statement of fact or declaration of purpose

Open values [11], [12]

- TBD = to be defined / determined · TBS = to be specified · TBR = to be resolved (NASA)
- Minimize their use and track every open value until it is closed

Terms not to be used (ECSS-E-ST-10-06C, cl. 8.3.3), e.g. [6]

- “and/or”, “etc.”, “goal”, “appropriate”, “as far as possible”, “optimize”, “minimize”, “maximize”, “typical”, “rapid”, “user-friendly”, “easy”, “sufficient”, “adequate”, “small”, “large”, “state of the art”

Examples: Poor vs. Improved Requirements

Rewriting requirements (illustrative values, not from a real mission)

Poor	Problem	Improved
The EPS shall provide sufficient power for the payload.	Unverifiable term (“sufficient”); no value, no condition	The EPS shall provide ≥ 60 W orbit-average power to the payload in Science Mode at EOL.
The AOCS shall point the camera accurately and quickly using reaction wheels.	Two requirements in one; “accurately”, “quickly” unverifiable; prescribes the solution (how)	(a) The AOCS shall point the camera boresight to within 0.1° (3σ) of the commanded attitude in Science Mode. (b) The AOCS shall perform a 30° slew, including settling, within 120 s.
The OBC shall store payload and/or housekeeping data etc.	“and/or”, “etc.”; no capacity, no duration	(a) The OBC shall store ≥ 16 GB of payload data. (b) The OBC shall store ≥ 72 h of housekeeping telemetry.

Verification of Requirements

Verification Methods [8]

- Verification: “Was the end product realized right?” · Validation: “Was the right end product realized?”

Method	Definition (ECSS-E-ST-10-02C Rev.1)	Typical evidence / example
Test (T) incl. demonstration	Measuring product performance and functions under representative simulated environments	e.g. thermal-vacuum or vibration test
Analysis (A) incl. similarity	Theoretical or empirical evaluation using techniques agreed with the customer	Modelling and computational simulation, e.g. pointing budget
Review-of-design (ROD)	Using approved records or evidence that unambiguously show that the requirement is met	Design documents and reports, technical descriptions, engineering drawings
Inspection (I)	Visual determination of physical characteristics	Constructional features, workmanship, code vs. coding standard

➔ Listed in order of precedence (in general more confidence).

Verification Levels, Stages and Verification Matrix [8]

Levels	Equipment	Subsystem	Element	Segment	Overall system
Stages	Qualification	Acceptance	Pre-launch	In-orbit (incl. commissioning)	Post-landing

Verification Control Document (VCD) = verification matrix, a living document

Req. ID	Requirement (short)	Method	Level	Stage	Status
SYS-EPS-030	≥ 60 W orbit-average payload power at EOL	A, T	System	Qualification	Open
AOCS-PER-010	Pointing ≤ 0.1° (3σ) in Science Mode	A	System	Qualification, in-orbit	Open
OBC-DAT-020	≥ 16 GB payload data storage	T	Equipment	Acceptance	Closed
STR-MEC-005	Connector types as specified in the ICD	I	Equipment	Acceptance	Closed

Per requirement the VCD also records traceability, compliance evidence and close-out status. Example entries are illustrative.

Lessons Learned: When Requirements Fail

Mission	What happened	Lesson for requirements
Mars Climate Orbiter NASA, 1999 [13]	Ground software file “Small Forces” delivered thruster impulse data in lbf·s instead of the N·s specified in the Software Interface Specification → navigation error, spacecraft lost at Mars orbit insertion	Specify interfaces completely (incl. units) and verify them end-to-end
Mars Polar Lander NASA, 1999 [14]	Probable cause: a spurious touchdown signal at landing-leg deployment – a known characteristic – was retained by the flight software → descent engines shut down at ~40 m altitude	Flow known hardware behaviour down into the software requirements; test in flight configuration
Ariane 5 Flight 501 1996 [15]	Inertial reference software re-used from Ariane 4; Ariane 5 trajectory data were not included in its requirements → conversion overflow, loss of guidance ~37 s after ignition	Re-validate requirements of re-used items: “Include trajectory data in specifications and test requirements” (R10)

Image index

[1] NASA Systems Engineering Handbook (NASA/SP-2007-6105), 2007

[2] https://www.engineering.com/story/model-based-system-engineering---beyond-spreadsheets

References

[3] J. M. Stecklein et al., “Error Cost Escalation Through the Project Life Cycle,” 14th Annual INCOSE International Symposium, Toulouse, June 2004. NTRS 20100036670

[4] ECSS-S-ST-00-01C Rev.1, Glossary of terms, 11 October 2023

[5] ECSS-E-ST-10C Rev.1, Space engineering – System engineering general requirements, 15 February 2017

[6] ECSS-E-ST-10-06C, Space engineering – Technical requirements specification, 6 March 2009

[7] ECSS-E-ST-10-24C Rev.1, Space engineering – Interface management, 15 November 2024

[8] ECSS-E-ST-10-02C Rev.1, Space engineering – Verification, 1 February 2018

[9] ESA, Margin philosophy for science assessment studies, SRE-PA/2011.097, Issue 1 Rev. 3, 15 June 2012

[10] ECSS-M-ST-10C Rev.1, Space project management – Project planning and implementation, 6 March 2009

[11] NASA Systems Engineering Handbook, NASA/SP-2016-6105 Rev2, 2016 (web edition nasa.gov/seh: Sec. 5.3, App. C)

[12] ECSS-E-ST-60-30C, Space engineering – Satellite attitude and orbit control system (AOCS) requirements, 30 August 2013

[13] Mars Climate Orbiter Mishap Investigation Board, Phase I Report, NASA, 10 November 1999

[14] JPL Special Review Board, Report on the Loss of the Mars Polar Lander and Deep Space 2 Missions, JPL D-18709, 22 March 2000

[15] Inquiry Board (chair J. L. Lions), ARIANE 5 Flight 501 Failure, Paris, 19 July 1996

[16] ESA, “ESA’s Hera targets early arrival at Didymos asteroids,” esa.int, 7 October 2025